



Yeshwantrao Chavan College of Engineering

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Department of Information Technology



Session 2026-27
Project Phase-II
Title Selection Seminar
on
Sovereign-RAG Omni: A Secure Offline Architecture for Multimodal Retrieval, Voice Interaction, and Automated Visual Synthesis

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Abstract

- Sovereign RAG is a privacy-first, offline Retrieval-Augmented Generation (RAG) system for secure document-based question answering.
- Uses local Large Language Models (LLMs) to retrieve and generate accurate, context-aware responses.
- Runs entirely on the user's system, ensuring 100% data privacy without relying on cloud APIs.
- Phase-I implemented document ingestion, semantic indexing, hybrid retrieval, and local response generation using Ollama.
- Phase-II extends the system with:
 - Multimodal document processing
 - Voice interaction
 - Intelligent image retrieval
 - Automatic mind map generation
 - Automatic flowchart generation
- Provides a secure, efficient, and user-friendly AI assistant for offline knowledge retrieval.

Introduction

Background

- Organizations generate large volumes of confidential documents, including reports, manuals, research papers, invoices, presentations, and meeting records.
- Cloud-based AI services require uploading these documents to external servers, leading to privacy and security concerns.
- Retrieval-Augmented Generation (RAG) combines document retrieval with Large Language Models (LLMs) to generate accurate, context-aware responses.
- Sovereign RAG performs the entire retrieval and response generation process locally, ensuring complete data privacy and eliminating internet dependency.
- The system provides a secure, offline AI solution for efficient knowledge retrieval from local documents.

Problem Definition

Existing Problems:

1. Sensitive organizational data is exposed when using cloud AI platforms.
2. Most offline RAG systems support only textual documents.
3. Images and audio recordings cannot be searched efficiently.
4. Long textual responses are difficult to understand.
5. Lack of voice interaction affects accessibility.
6. Existing systems provide limited visualization of retrieved knowledge.

Aim and Objectives

Aim:

To develop a privacy-preserving offline multimodal Retrieval-Augmented Generation system capable of retrieving information from text, audio, and images while providing intelligent voice interaction and visual knowledge representation.

Objectives:

- Develop a completely offline AI assistant.
- Implement efficient PDF and document ingestion.
- Improve retrieval using Hybrid Search.
- Integrate Voice Assistant using Speech-to-Text and Text-to-Speech.
- Enable Audio File Processing.
- Implement Semantic Image Search.
- Generate Mind Maps automatically.
- Generate Flowcharts from retrieved knowledge.
- Maintain complete data privacy.

Literature Survey

Sr. No.	Title of the Paper	Author	Year of Publication / Publisher	Major Observations / Findings
1	<i>Ask in Any Modality: A Comprehensive Survey on Multimodal Retrieval-</i>	Mohammad Mahdi Abootorabi et al.	2025 / ACL Findings	Provides a comprehensive survey of Multimodal RAG, covering text, image, audio, and video retrieval, along with challenges and future research directions. (ACL Anthology)
2	<i>Multimodal Retrieval-Augmented Generation Framework for Visually Rich Knowledge in the</i>	Xianchuan Meng, Ziyu Tong	2025 / Springer (Architectural Intelligence)	Proposes a multimodal RAG framework for processing visually rich documents, improving retrieval and decision-making using text and image information. (Springer)
3	<i>A Survey of Multimodal Retrieval-Augmented Generation</i>	Lang Mei, Siyu Mo, Zhihan Yang, Chong Chen	2025 / arXiv	Reviews recent Multimodal RAG techniques, including retrieval, fusion, generation, datasets, benchmarks, and evaluation methods for multimodal AI systems. (arxiv.org)

Patent Search

Sr. No.	Title of the Patent	Author	Year of Publication/ Publisher	Major observations/findings
1.	Response Generation Using RAG US2024346256 (A1)	Qin Yinghua [US]	2024-10-17	It accepts voice queries, converts them to text, retrieves relevant information, and generates a response.
2.	Domain Adaptation of ASR with RAG	Chauhan Sanjay[IN], Jain Mayank[IN]	2024-07-12	Accepts image + text queries together, creates joint embeddings, retrieves passages, and generates grounded answers.
3.	Multimodal Retrieval- Augmented Visual QA	Khare Anuj [IN]	2024-06-26	Optimizes ingestion/retrieval for mixed media datasets (documents with text + graphics).

Research Gap

Existing Systems

- Depend on cloud APIs.
- Limited data privacy.
- Mostly text-based.
- Lack multimodal support.
- Poor visualization.

Proposed Solution Sovereign RAG solves these problems by

- Complete offline execution
- Local LLMs
- Hybrid Search
- Voice Assistant
- Audio Processing
- Image Retrieval
- Mind Map Generation
- Flowchart Generation

Hardware/Software Requirement

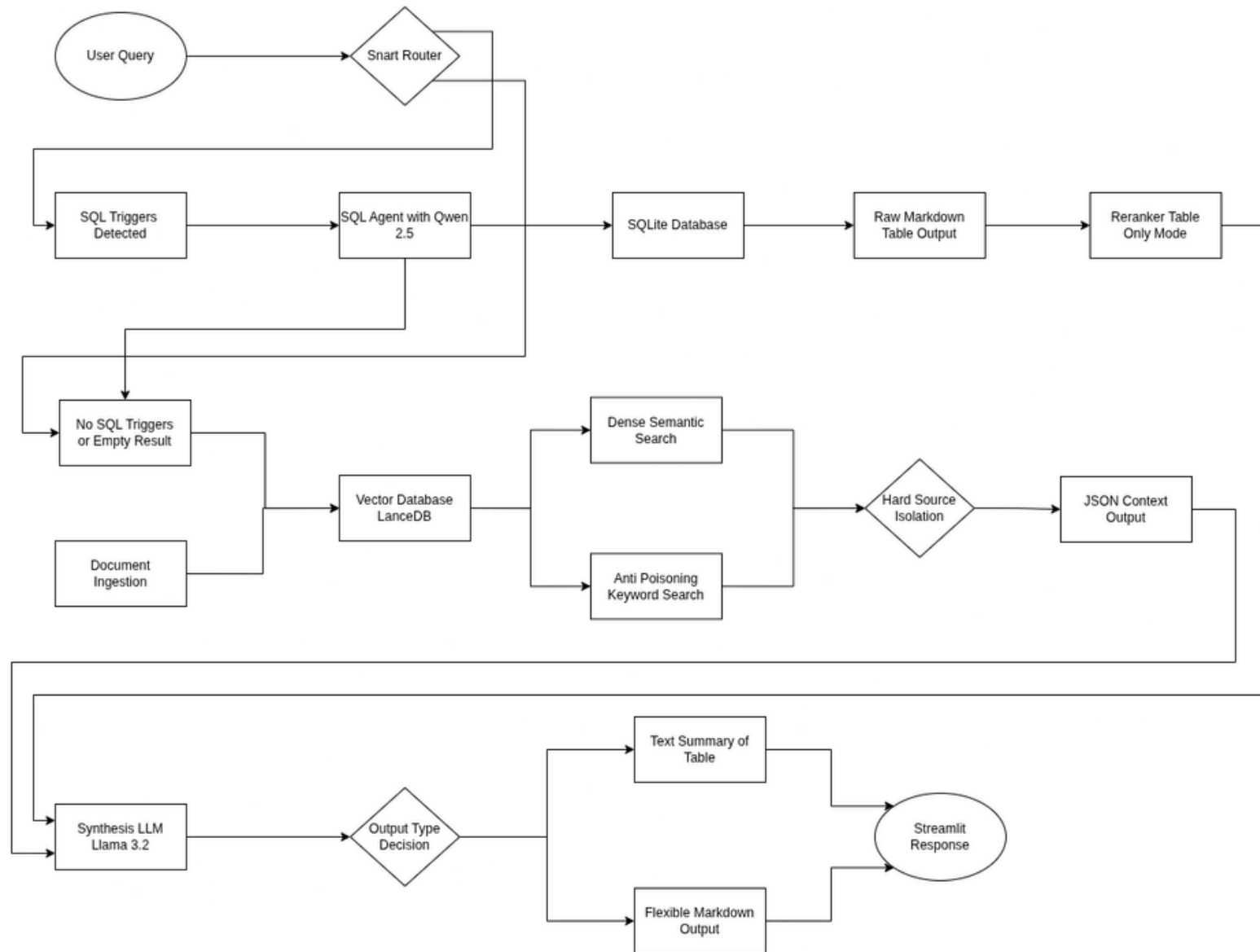
Software

- Python
- Streamlit
- LangChain
- Ollama
- ChromaDB
- PyMuPDF
- MarkdownIt
- Pillow
- Faster Whisper
- CLIP
- Mermaid.js

Hardware

- Intel i5 / Ryzen 5+
- 16 GB RAM
- SSD Storage
- NVIDIA GPU (Recommended)
- Windows / Linux

Proposed Methodology Phase-I



Experimental results/Snapshots Phase I

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Deploy 



Pressure conversions



✓  Complete

To convert between different units of pressure, we can use the following formulas:

- 1 bar = 1.01325 atm
- 1 psi ≈ 0.07031 bar
- 1 N/mm² ≈ 1.0197 bar (or approximately 14.503 psi)
- 1 MPa ≈ 145.03 psi

Please note that these values are provided on Page 21 of the manual, Fig. 2.2 Pressure conversions.

When working with pressure gauges, it is essential to understand the difference between absolute and relative pressures. Absolute pressure measures pressure from a reference point (usually vacuum), while relative pressure measures pressure above atmospheric pressure.

In the context of hydraulic systems, pressure in the system can change from inlet to outlet due to friction losses and other factors. The formula for calculating hydraulic power is:

Output Cylinder Power = Force × Speed = (Pressure × Cylinder Area) × (Inlet Flow Rate / Cylinder Area)

This formula is provided on Page 802 of the manual.

When working with seals in hydraulic systems, it is crucial to ensure that they are compatible with the fluid being used and that the pressure rating is adequate. Seals should be used for atmospheric pressure on air sides and sealing fluids at pressures above 0 kPa (0 bar).

Please cite the page numbers when necessary, but as this information was provided directly in the question context, there is no need to do so.

>  View Source References

Ask about your documents...



Experimental results/Snapshots Phase I

>>

Deploy ⋮

PDFs

Spreadsheets

Word Docs

 show records on date 12-12-2024 from maintenance_history_1000.csv  Complete

Retrieved Records:

ticket_id	date	machine_type	reported_fault	technician	downtime_hours	resolution_notes
TKT-1870	2024-12-12	DC Electric Motor	Rotor jammed	Carlos Mendez	13.8	Applied Fleming's Left Hand Rule to verify field orientation. Replaced worn carbon commutator brushes.

🗨️ **AI Analysis: Interpretation:** You asked for the records from a maintenance history database, specifically focusing on data related to DC Electric Motors on December 12th, 2024.

The analysis of the provided records reveals that **one incident was reported**, involving a DC Electric Motor with a rotor jam issue. The details of this incident are as follows:

- Machine Type: DC Electric Motor
- Reported Fault: Rotor jammed
- Technician: Carlos Mendez
- Resolution Notes: • Applied Fleming's Left Hand Rule to verify field orientation. • Replaced worn carbon commutator brushes.

There are no other records or incidents reported on this date, making it a singular event in the maintenance history database.

Ask about your documents...



Experimental results/Snapshots Phase I

Deploy

Document Manager

☐ Enable folder selection

Upload Documents or Spreadsheets

Drag and drop files here

Limit 200MB per file • PDF, CSV, XLSX,...

Browse files

 Standard Operations...
3.0MB

 maintenance_histor...
159.7KB

 Maintenance of Mac...
1.8MB

Showing page 1 of 2



System Health & Debug

✓ DB Connected: 8803
chunks

 Full Reset

- Synthetic base stock

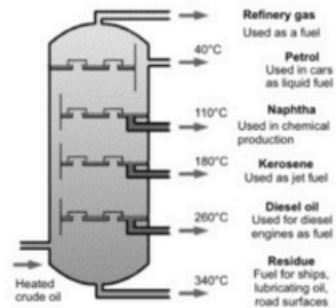
Crude oil consists of a complex of many different hydrocarbon compounds of various weights and densities, each having a different boiling point.

Physical properties of the oil depend upon the base stock.

References:

- Page 739: DIAGRAM CAPTION
- Page 742 (not shown in excerpt): Further information on Base Stocks-Mineral & Other Oils (11.3.1)
- OSHA regulations, Section 3.2, Page 1024

by erosion. Figure 11.5 shows the refining of crude petroleum oil and the different outputs.



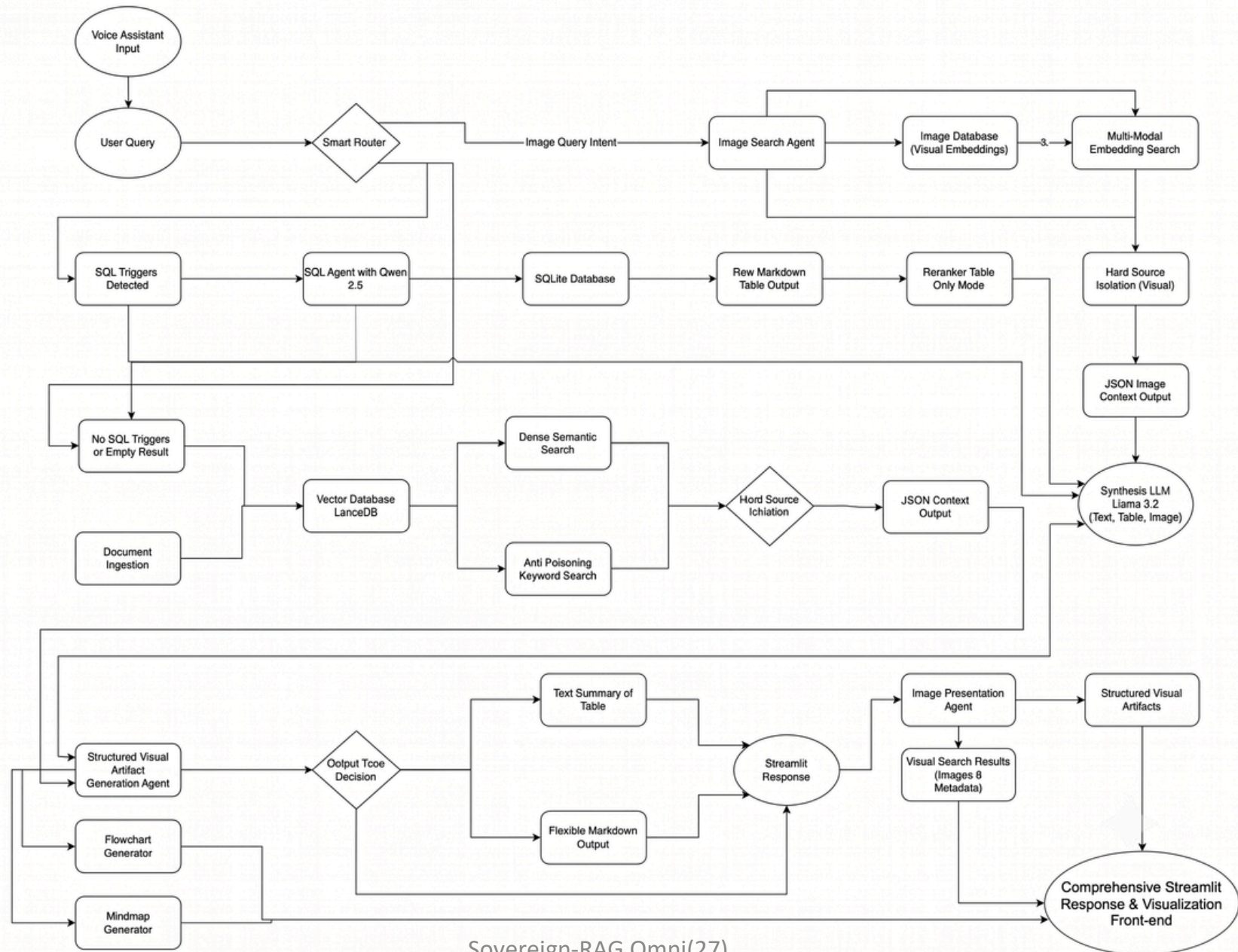
by erosion. Figure 11.5 shows the refining of crude petroleum oil and the different

>  View Source References

Ask about your documents...



Proposed Methodology Phase-II



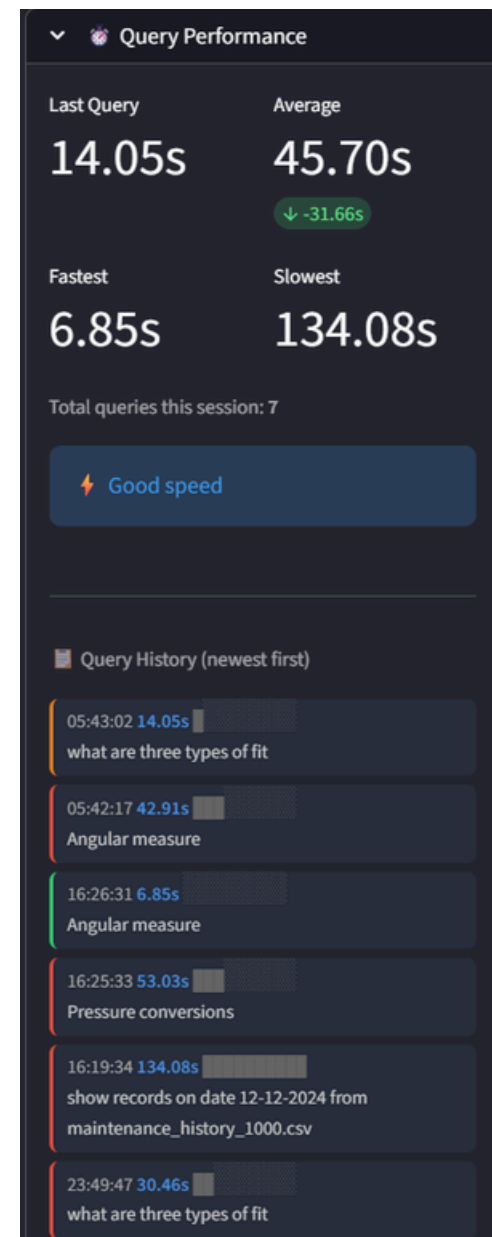
Analysis of Result

Phase-I Performance & Limitations

- Privacy & Security: 100% offline processing with zero external API calls, ensuring complete data privacy.
- Performance: Average query response time of 40-50 seconds due to local LLM inference and vector retrieval.
- Current Limitation: Supports only text-based queries and responses, making it time-consuming to understand lengthy outputs.

Phase-II Enhancements

- Voice Interaction: Offline Speech-to-Text (STT) and Text-to-Speech (TTS) for hands-free communication.
- Audio Ingestion: Indexing and retrieval from audio recordings and lectures.
- Intelligent Image Retrieval: Semantic search across local images using multimodal embeddings.
- Knowledge Visualization: Automatic generation of mind maps and flowcharts for faster understanding.



Plan of Action					
Activity (2025)	July-Aug	Aug-Sept	Sept-Oct	Oct-Nov	Nov-Dec
Literature Survey					
Analysis of literature data					
Conclusions					
Preparation of Project Report					

References

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- [9] A. Radford et al., "Robust Speech Recognition via Large-Scale Weak Supervision," in *Proc. International Conference on Machine Learning (ICML)*, 2023, pp. 28492–28518.
- [10] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 9459–9474.

Roles and Responsibility

Team member Details	Roles and Responsibility
Amruta Thakur(01)	focus on LLM integration, Gemini vs RAG comparison, hallucination reduction, and evaluation framework.
Nishant Lilhare(58)	design and implement the core Offline RAG pipeline, vector database integration, and overall system architecture.
Riya Kawitkar (109)	develop the Streamlit dashboard, visualization module, comparison UI, and assist in testing and result analysis
Krishna Agrawal(137)	literature survey, problem definition, research gap identification, and preparation of abstract, PPT, and final report
Swaraj Rohad(155)	work on data ingestion modules (PDF, CSV, Images), chunking strategies, and preprocessing for multimodal documents.

Thank You.....

Proposed Methodology Phase-II

